# K-Means Clustering Assignment

## Dataset

Mall Customer Segmentation Data (Kaggle)

Link: https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python

## Objective

Perform customer segmentation using K-Means Clustering to identify distinct customer groups based on their purchasing patterns and demographics.

## Tasks

- 1. Load and Explore the Dataset:
  - Import the dataset and display the first few rows.
  - Provide a summary of key features (count, mean, std, min, max).
- 2. Preprocess the Data:
  - Check and handle missing values.
  - Encode categorical variables (if any).
  - Normalize or scale numerical features if needed.
- 3. Feature Selection:
  - Select the most relevant features (e.g., Annual Income, Spending Score).
  - Justify why you selected those features for clustering.
- 4. Determine the Optimal Number of Clusters (k):
  - Use the Elbow Method to plot WCSS (Within-Cluster Sum of Squares) vs. k.
  - Identify the 'elbow point' for the ideal number of clusters.
- 5. Apply K-Means Clustering:
  - Fit the K-Means model with the chosen k.
  - Assign cluster labels to each data point.
  - Visualize the clusters using a scatter plot (e.g., Annual Income vs. Spending Score).
- 6. Interpret Clusters:
  - Analyze each cluster and describe their behavior in business terms (e.g., 'High income, low spending').

## Scenario-Based Questions

### A. Marketing Strategy

You have identified clusters of customers. For each cluster, propose a targeted marketing strategy. Which cluster would be most appropriate for premium product targeting, and why?

- Sample Input:

| CustomerID | Age | Annual Income (k$) | Spending Score (1–100) | Cluster |
|---|---|---|---|---|
| 12 | 26 | 78 | 85 | 2 |
| 45 | 41 | 120 | 40 | 3 |

- Sample Output:

Cluster 2 – Young, high-spending customers: Target with luxury product ads and exclusive deals.
Cluster 3 – Middle-aged, high-income, low-spending: Offer premium subscription or loyalty benefits.

### B. Product/Service Design

Your company plans to launch a new loyalty program or customized service. Based on the cluster profiles, identify which clusters should be prioritized for this initiative and design at least two personalized offers or services per cluster.

- Sample Input:

Cluster Summary:
- Cluster 0: Low Income – Low Spending
- Cluster 1: High Income – High Spending
- Cluster 2: Middle Income – Moderate Spending

- Sample Output:

- Cluster 1 (High Income – High Spending):
  - Personalized luxury offers (early access to new products)
  - VIP membership program with cashback rewards

- Cluster 2 (Middle Income – Moderate Spending):
  - Discount coupons for repeat purchases
  - Loyalty points redeemable for free delivery